Quiz app

The Quiz App is a mini interactive web application developed using HTML, CSS, and JavaScript. The main objective of this project is to demonstrate the use of JavaScript DOM manipulation to create a dynamic and user-friendly quiz system.

The application displays multiple-choice questions one at a time, allows users to select answers, and includes a timer for each question. At the end of the quiz, the system automatically calculates and displays the final score and percentage. This project helps in understanding core JavaScript concepts such as event handling, conditional logic, and real-time UI updates

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Advanced Quiz App</title>


<style>

  * {

    box-sizing: border-box;

  }


  body {     margin: 0;     font-family: Arial, sans-serif;

background: linear-gradient(to right, #667eea, #764ba2);

height: 100vh;

    display: flex; justify-content:

    center;     align-items: center;

  }
```

```
  .quiz-box {     background: #ffffff;     width:
420px;     padding: 25px;     border-radius:
15px;     box-shadow: 0 10px 30px
rgba(0,0,0,0.3);
  }

  .timer {     text-align:
right;     font-weight:
bold;     color: red;
margin-bottom: 10px;
  }

  h2 {     color: #000000;
font-size: 20px;
margin-bottom: 15px;
  }

  ul {
    list-style: none;
padding: 0;
  }
   li
{
    background: #f1f1f1;
margin: 10px 0;     padding:
12px;     border-radius: 8px;
  }
```

```
  label {     color:
#000000;     font-
size: 16px;
cursor: pointer;
  }

  input[type="radio"] {     margin-
right: 10px;
  }

  button {     width: 100%;
padding: 12px;
background: #4a90e2;
    color: #ffffff;
border: none;
border-radius: 8px;
font-size: 16px; cursor:
pointer; margin-top:
15px;

  }

  button:hover {
background: #357ab8;
  }
```

```
  .result {     text-
align: center;
  }

  .result h1 {
color: #27ae60;
  }
</style>
</head>

<body>

<div class="quiz-box" id="quiz">
  <div class="timer" id="timer">Time: 30s</div>

  <h2 id="question">Question text will appear here</h2>

  <ul>
    <li>
      <input type="radio" name="answer" id="a" class="answer">
      <label for="a" id="a_text">Option A</label>
    </li>
    <li>
      <input type="radio" name="answer" id="b" class="answer">
      <label for="b" id="b_text">Option B</label>
    </li>
    <li>
```

```
      <input type="radio" name="answer" id="c" class="answer">
      <label for="c" id="c_text">Option C</label>
    </li>
    <li>
      <input type="radio" name="answer" id="d" class="answer">
      <label for="d" id="d_text">Option D</label>
    </li>
  </ul>

  <button id="submit">Next</button>
</div>

<script>   const
quizData = [
    {
      question: "Which language runs in the browser?",
a: "Java",      b: "C",      c: "Python",      d: "JavaScript",
correct: "d"
    },
    {
      question: "What does DOM stand for?",
a: "Data Object Model",      b: "Document
Object Model",      c: "Digital Output
Method",      d: "Desktop Oriented Mode",
correct: "b"
    },
    {
```

```
    question: "Which keyword is used to declare variables in JavaScript?",
a: "int",      b: "string",      c: "var",      d: "define",      correct: "c"
   }
 ];

 const quiz = document.getElementById("quiz");   const
questionEl = document.getElementById("question");   const
answerEls = document.querySelectorAll(".answer");   const
timerEl = document.getElementById("timer");   const
submitBtn = document.getElementById("submit");

 const a_text = document.getElementById("a_text");   const
b_text = document.getElementById("b_text");   const c_text
= document.getElementById("c_text");   const d_text =
document.getElementById("d_text");   let currentQuiz = 0;
let score = 0;   let timeLeft = 30;   let timer;

 loadQuiz();
startTimer();

 function loadQuiz() {
deselectAnswers();     timeLeft
= 30;

   const q = quizData[currentQuiz];
questionEl.innerText = q.question;
a_text.innerText = q.a;     b_text.innerText
```

```
= q.b;     c_text.innerText = q.c;
d_text.innerText = q.d;
  }

  function deselectAnswers() {     answerEls.forEach(el
=> el.checked = false);
  }

  function getSelected() {
let answer;
answerEls.forEach(el => {
if (el.checked) {

        answer = el.id;
      }
});
    return answer;
  }

  function startTimer() {     timer = setInterval(()
=> {       timerEl.innerText = "Time: " + timeLeft +
"s";       timeLeft--;

      if (timeLeft < 0) {
nextQuestion();
      }
    }, 1000);
  }
```

```
  function nextQuestion() {
clearInterval(timer);

    const answer = getSelected();     if (answer ===
quizData[currentQuiz].correct) {       score++;
    }

    currentQuiz++;

    if (currentQuiz < quizData.length) {
      loadQuiz();
startTimer();
    } else {
showResult();
    }
  }

  submitBtn.addEventListener("click", nextQuestion);

  function showResult() {     const percentage = Math.round((score /
quizData.length) * 100);     quiz.innerHTML = `
      <div class="result">
        <h1>Quiz Completed  </h1>
        <p>Score: ${score} / ${quizData.length}</p>
        <p>Percentage: ${percentage}%</p>
        <button onclick="location.reload()">Restart Quiz</button>
```

```
    </div>
  `;
 }
</script>


</body>
</html>
```

Time: 24s

What does DOM stand for?

- Data Object Model
- Document Object Model
- Digital Output Method
- Desktop Oriented Mode

Next

Quiz Completed 🎉

Score: 2 / 3

Percentage: 67%

Restart Quiz